

Robotics Applications in Healthcare: Surgical Robots and Rehabilitation Systems

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Abstract

This paper presents a comprehensive review of robotics applications in healthcare, with a specific focus on surgical robotic systems and rehabilitation robotics. Advances in mechanical design, artificial intelligence (AI), computer vision, and human-machine interaction have accelerated the integration of robotic systems into clinical environments. Surgical robots such as the da Vinci Surgical System have transformed minimally invasive surgery by offering enhanced precision, tremor elimination, and three-dimensional visualization. Concurrently, rehabilitation robots including exoskeletons and assistive devices have demonstrated significant potential in restoring motor function in patients suffering from stroke, spinal cord injury, and neurodegenerative diseases. This review covers the fundamental principles underlying these systems, examines key technologies and clinical outcomes, identifies current limitations, and outlines future research directions. The paper concludes that despite regulatory and economic challenges, robotic healthcare technologies hold transformative potential for improving patient outcomes and redefining surgical and therapeutic standards worldwide.

I. INTRODUCTION

The field of medical robotics has undergone remarkable transformation over the past three decades. What began as laboratory demonstrations of teleoperated manipulators has evolved into sophisticated, FDA approved clinical platforms that are deployed in thousands of hospitals worldwide. The global medical robotics market was valued at approximately USD 14.7 billion in 2023 and is projected to exceed USD 45 billion by 2030, growing at a compound annual growth rate (CAGR) of over 17% [1].

Robotics in healthcare can be broadly classified into two principal domains: (1) surgical robotics, which assists or augments surgeons during operative procedures, (2) rehabilitation robotics, which aids patients in recovering physical function following neurological or musculoskeletal impairment. Both domains leverage converging technologies including mechatronics, artificial intelligence, sensor fusion, and advanced materials to deliver measurable clinical benefits.

The motivation for robotic assistance in surgery is compelling. Human surgical performance is inherently limited by physiological tremor, ergonomic fatigue, restricted field of view, and the challenge of manipulating delicate tissue in constrained anatomical spaces. Robotic systems overcome many of these constraints by translating gross hand movements into micro-scale, tremor free manipulations with haptic scaling and real-time imaging integration [2].

Similarly, neurological rehabilitation has traditionally relied on labor intensive, therapist administered repetitive exercises. Rehabilitation robots offer the prospect of high dose, precisely quantified, and patient adaptive therapy that can be delivered consistently and at reduced cost. Devices such as the Lokomat gait trainer, Ekso Bionics exoskeleton, and MIT Manus upper-limb robot have been validated in randomized controlled trials [3].

This paper is organized as follows: Section II reviews surgical robotic systems; Section III discusses rehabilitation robotics; Section IV examines enabling technologies; Section V presents clinical outcomes and comparative analysis; Section VI identifies limitations and ethical considerations; and Section VII concludes with future perspectives.

II. SURGICAL ROBOTIC SYSTEMS

A. Historical Development

The genesis of surgical robotics can be traced to the ROBODOC system, introduced in 1992 for computer assisted orthopedic surgery, and to AESOP (Automated Endoscopic System for Optimal Positioning), which performed automated camera navigation in laparoscopic procedures [4]. The landmark development came in 2000 when Intuitive Surgical received FDA clearance for the da Vinci Surgical System, marking the beginning of robot-assisted minimally invasive surgery (RMIS) as a commercial and clinical reality.

B. The da Vinci Surgical System

The da Vinci system represents the current gold standard in robotic surgery. The fourth generation da Vinci Xi platform consists of three primary components: (1) a surgeon console with a stereoscopic viewer and master controls; (2) a patient side cart with four robotic arms; and (3) a vision cart housing the image processing and electrosurgical hardware [5]. The system uses a master-slave control architecture in which the surgeon's hand movements are digitized, scaled (up to 5:1 motion scaling), and filtered to eliminate physiological tremor before being transmitted to the EndoWrist instruments.

EndoWrist instruments provide seven degrees of freedom (DOF) within the patient's body cavity surpassing the six DOF of the human wrist and achieve angular ranges of up to 540° rotation. The three-dimensional, high-definition visualization system provides 10× magnification, enabling surgeons to operate within anatomical regions inaccessible by conventional laparoscopy. Over 10 million da Vinci procedures have been performed globally as of 2023, spanning prostatectomy, hysterectomy, cardiac valve repair, thoracic surgery, and colorectal resection [6].

C. Alternative Platforms and Emerging Systems

Competition and innovation have produced several alternative surgical robotic platforms. The Versius system (CMR Surgical) employs a modular, portable design with lightweight robotic arms that can be repositioned rapidly between procedures. The Hugo RAS system (Medtronic) features an open console architecture intended to improve surgical team communication. The SPORT Surgical System (Titan Medical) is being developed as a single port robotic platform [7].

Autonomous and semi-autonomous surgical robots represent the next frontier. The Smart Tissue Autonomous Robot (STAR) has demonstrated the capacity to perform supervised autonomous suturing of soft tissue with outcomes superior to human surgeons in porcine models, as published in Science Robotics in 2022 [8]. Robotic systems for neurosurgery, ophthalmic surgery, and intravascular procedures are in various stages of clinical translation.

III. REHABILITATION ROBOTICS

A. Rationale and Neuroplasticity

Neurological rehabilitation exploits the brain's capacity for neuroplasticity the ability to reorganize synaptic connections in response to activity and learning. Evidence from motor learning research indicates that high repetition, task specific training drives cortical remapping and functional recovery after stroke or spinal cord injury [9]. Conventional therapy is constrained by therapist bandwidth and patient fatigue; robotic systems overcome these barriers by delivering hundreds of repetitions per session with adaptive challenge modulation.

B. Upper-Limb Rehabilitation Robots

The MIT-Manus (InMotion ARM) is among the most extensively studied upper-limb rehabilitation robots. It is an impedance-controlled, two-degree of freedom planar robot that guides or resists shoulder and elbow movements during reaching tasks. A landmark multicenter randomized controlled trial demonstrated that robot assisted upper-limb therapy produced significantly greater motor gains than usual care in chronic stroke survivors [10]. The Armeo Spring (Hocoma) and Armeo Power exoskeletons extend the kinematic workspace to include the forearm, wrist, and hand, enabling task-oriented virtual reality therapy.

C. Lower-Limb and Gait Rehabilitation

The Lokomat (Hocoma AG) is the most widely deployed lower-limb rehabilitation robot. It consists of a motorized exoskeleton worn on the legs, suspended over a body weight support treadmill. Actuated hip and knee joints guide the patient through physiologically accurate gait cycles, while an algorithm adjusts assistance levels based on measured joint torque. A systematic review and meta-analysis of 22 randomized controlled trials found that Lokomat training significantly improved walking speed, endurance, and balance in stroke and spinal cord injury patients compared to conventional physiotherapy [11].

Wearable exoskeletons for over-ground ambulation include the Ekso GT (Ekso Bionics) and ReWalk Personal 6.0 (ReWalk Robotics), both cleared by the FDA for clinical and personal use in spinal cord injury. These systems use inertial measurement units (IMUs) and force sensors to detect user intent and trigger powered hip and knee flexion extension cycles, enabling individuals with complete motor paraplegia to achieve over ground walking [12].

D. Socially Assistive and Cognitive Rehabilitation Robots

Beyond motor rehabilitation, robots such as PARO (a therapeutic robotic seal) have been used in dementia and psychiatric care to reduce agitation and improve social engagement. Humanoid robots including NAO and Pepper have been deployed in autism spectrum disorder (ASD) therapy, where their predictable and consistent behavior helps children practice social interaction skills in a low-anxiety environment [13].

IV. ENABLING TECHNOLOGIES

A. Artificial Intelligence and Machine Learning

AI and machine learning are increasingly integral to both surgical and rehabilitation robotics. In surgical systems, convolutional neural networks (CNNs) analyze endoscopic video to perform real time tissue segmentation, instrument tracking, and critical structure identification (e.g., ureter or bile duct

detection to prevent inadvertent injury) [14]. Reinforcement learning (RL) has been applied to train autonomous surgical subtask policies, including needle grasping, suture threading, and knot tying.

In rehabilitation, adaptive control algorithms modulate robotic assistance based on continuously estimated user performance metrics. Electromyography (EMG) driven control interfaces decode intended limb movement from residual muscle signals, enabling robots to move in synchrony with the user's voluntary effort a key requirement for promoting neuroplasticity [15].

B. Human-Machine Interaction and Haptics

Force feedback (haptics) remains a critical and partially unsolved challenge in surgical robotics. Current commercial systems such as the da Vinci provide no haptic feedback to the surgeon, relying instead on visual force cues. Research prototypes employing piezoelectric actuators and magnetorheological fluid transmissions have demonstrated tactile rendering at surgical scales, but translation to clinical practice faces barriers of cost, sterilization compatibility, and latency [16].

C. Imaging Integration and Navigation

Intraoperative image guidance is a core enabling technology. Robotic systems increasingly incorporate fluoroscopy, ultrasound, and magnetic resonance imaging (MRI) compatibility to enable real time navigation. The Mazor X Stealth Edition (Medtronic) fuses preoperative CT planning with intraoperative imaging to guide robotic arm positioning for spinal pedicle screw placement with sub millimeter accuracy [17].

V. CLINICAL OUTCOMES AND COMPARATIVE ANALYSIS

Application	System	Key Metric	Outcome vs. Control
Prostatectomy	da Vinci	Blood loss	Down 50-70% vs. open surgery
Stroke - Upper Limb	MIT-Manus	Fugl-Meyer score	+4.5 pts vs. usual care
SCI Gait Training	Lokomat	10-m walk speed	+0.12 m/s vs. conventional PT
Spinal Surgery	Mazor X	Screw accuracy	99.1% grade A vs. 91.4% freehand
Cardiac Valve Repair	da Vinci	Hospital stay	Down 2-3 days vs. sternotomy

TABLE I. SUMMARY OF CLINICAL OUTCOMES FOR ROBOTIC HEALTHCARE SYSTEMS

Robot-assisted radical prostatectomy (RARP) has been the most extensively evaluated RMIS application. A meta-analysis of 57 studies encompassing over 400,000 patients found that RARP was associated with significantly lower estimated blood loss, reduced transfusion rates, shorter hospital stay, and lower incidence of postoperative complications compared to open and laparoscopic approaches, with equivalent oncologic outcomes at 5-year follow-up [18].

For rehabilitation, effect sizes vary by patient population, timing of intervention, and device type. A 2023 Cochrane review of robot assisted gait training post stroke found moderate certainty evidence that robotic gait training combined with physiotherapy increased the likelihood of independent walking compared to physiotherapy alone (RR 1.21, 95% CI 1.06–1.38) [19].

VI. LIMITATIONS, CHALLENGES, AND ETHICAL CONSIDERATIONS

A. Technical Limitations

Current surgical robotic systems exhibit several technical deficiencies. The absence of haptic feedback represents a significant safety concern; surgeons report difficulty gauging tissue tension, which can lead to inadvertent suture breakage or tissue tearing. System latency, while typically under 50 ms in current platforms, becomes clinically significant in procedures requiring rapid force modulation. Robotic arm size and limited workspace maneuverability constrain applicability in narrow anatomical regions [20].

B. Economic and Access Barriers

The capital cost of a da Vinci Xi system exceeds USD 2 million, with annual service contracts of USD 100,000–180,000 and per procedure instrument costs of USD 700–3,500. These economics restrict adoption to high volume centers in high income countries. A global analysis found that over 85% of installed surgical robots are located in North America, Western Europe, and East Asia, creating a stark disparity in access for patients in low and middle-income countries (LMICs) [21].

C. Regulatory and Liability Framework

The regulatory classification of increasingly autonomous robotic systems challenges existing frameworks. Current FDA guidance treats surgical robots as Class II or III devices requiring 510(k) or PMA clearance; however, AI enabled adaptive systems may require post market surveillance mechanisms that evolve with software updates. The question of liability allocation when an autonomous surgical action results in patient harm between the manufacturer, institution, and surgeon remains legally unresolved in most jurisdictions [22].

D. Ethical Considerations

The deployment of robotic systems raises important ethical questions regarding informed consent, data privacy, de-skilling of surgeons, and the appropriate boundary between machine autonomy and human oversight. Patients must be informed when robotic systems will be used in their care, and institutional review processes must evaluate AI decision making transparency. Rehabilitation robot developers must also address issues of user dignity, especially for elderly or cognitively impaired populations [23].

VII. FUTURE PERSPECTIVES

The trajectory of healthcare robotics points toward several transformative developments. Miniaturized microrobots, navigated by external magnetic fields, are being developed for targeted drug delivery, thrombus dissolution, and retinal microsurgery. Soft robotics systems constructed from compliant, deformable materials promise safer human-robot physical interaction and are particularly relevant for wearable rehabilitation devices and endoluminal surgical systems [24].

The convergence of 5G telecommunications with cloud robotics may enable remote tele-surgery over geographic distances, extending specialist surgical access to rural and underserved communities. Early demonstrations, including the remote laparoscopic procedure performed by Chinese surgeons over a 5G network in 2019, established proof of concept, though latency, cybersecurity, and regulatory approval remain barriers [25].

Brain computer interfaces (BCIs) integrated with rehabilitation exoskeletons represent a paradigm shift for spinal cord injury rehabilitation. Systems that decode motor intent from electroencephalography (EEG) or electrocorticography (ECoG) signals and actuate exoskeletal joints in closed loop have

demonstrated cortical reorganization consistent with neuroplastic recovery in early clinical trials. The integration of functional electrical stimulation (FES) with robotic exoskeletons offers the prospect of hybrid systems that engage both neural and mechanical pathways simultaneously [26].

VIII. CONCLUSION

This paper has reviewed the state of the art in healthcare robotics across surgical and rehabilitation domains. Surgical robots, led by the da Vinci platform, have achieved substantial clinical adoption and demonstrated measurable improvements in perioperative outcomes for a range of procedures. Rehabilitation robots, including stationary platforms and wearable exoskeletons, have produced evidence-based gains in motor function recovery for stroke and spinal cord injury populations.

The enabling technologies of AI, advanced sensing, human-machine interaction, and imaging integration are accelerating the capabilities of robotic healthcare systems. However, persistent challenges including high cost, limited haptic feedback, regulatory uncertainty, and inequitable global access must be systematically addressed to realize the full potential of these technologies.

Future research should prioritize cost reduction through platform commoditization, development of validated autonomy frameworks, and rigorous randomized controlled trials that establish long term efficacy and cost effectiveness. The integration of robotics with digital health infrastructure, AI driven personalization, and next generation neural interfaces will define the next chapter of medical robotics, offering the prospect of surgery and rehabilitation that is safer, more effective, and more accessible than ever before.

REFERENCES

- [1] Grand View Research, "Medical Robotics Market Size, Share & Trends Analysis Report," GVR-4-68039-074-5, San Francisco, CA, 2024.
- [2] R. H. Taylor, A. Menciassi, G. Fichtinger, P. Fiorini, and P. Dario, "Medical Robotics and Computer-Integrated Surgery," in *Springer Handbook of Robotics*, B. Siciliano and O. Khatib, Eds., Springer, Cham, 2016, pp. 1657–1684.
- [3] S. Hesse et al., "Treadmill training with partial body weight support compared with physiotherapy in nonambulatory hemiparetic patients," *Stroke*, vol. 26, no. 6, pp. 976–981, Jun. 1995.
- [4] W. Sackier and Y. Wang, "Robotically assisted laparoscopic surgery: From concept to development," *Surgical Endoscopy*, vol. 8, no. 1, pp. 63–66, Jan. 1994.
- [5] Intuitive Surgical Inc., "da Vinci Xi System: Product Technical Specifications," Sunnyvale, CA, USA, Rev. 5, 2022.
- [6] Intuitive Surgical Inc., "Annual Report 2023," Sunnyvale, CA, USA, 2024.
- [7] M. H. Hagen et al., "Robotic surgery in urology: Current and future perspectives," *Nature Reviews Urology*, vol. 20, pp. 481–495, Aug. 2023.
- [8] A. Shademan et al., "Supervised autonomous robotic soft tissue surgery," *Science Translational Medicine*, vol. 8, no. 337, Art. no. 337ra64, May 2016.
- [9] M. Lotze and U. Halsband, "Motor imagery," *Journal of Physiology*, vol. 99, no. 4–6, pp. 386–395, Jul.–Sep. 2006.
- [10] H. I. Krebs et al., "Increasing productivity and quality of care: Robot-aided neuro-rehabilitation," *Journal of Rehabilitation Research and Development*, vol. 37, no. 6, pp. 639–652, Nov.–Dec. 2000.

- [11] L. Mehrholz et al., "Electromechanical-assisted training for walking after stroke," *Cochrane Database of Systematic Reviews*, vol. 2017, no. 5, Art. no. CD006185, May 2017.
- [12] A. J. Esquenazi et al., "The ReWalk powered exoskeleton to restore ambulatory function to individuals with thoracic-level motor-complete spinal cord injury," *American Journal of Physical Medicine & Rehabilitation*, vol. 91, no. 11, pp. 911–921, Nov. 2012.
- [13] B. Scassellati, H. Admoni, and M. Mataric, "Robots for use in autism research," *Annual Review of Biomedical Engineering*, vol. 14, pp. 275–294, Aug. 2012.
- [14] D. Bouget et al., "Vision-based and marker-less surgical tool detection and tracking: A review of the literature," *Medical Image Analysis*, vol. 35, pp. 633–654, Jan. 2017.
- [15] E. Scheme and K. Englehart, "Electromyogram pattern recognition for control of powered upper-limb prostheses," *Journal of Rehabilitation Research and Development*, vol. 48, no. 6, pp. 643–660, 2011.
- [16] K. Leibrandt et al., "Implicit active constraints in robotic surgery for ensuring safe execution of learned trajectories," in *Proc. IEEE/RSJ IROS*, Chicago, IL, 2014, pp. 4331–4338.
- [17] F. Galbusera, G. Casaroli, and T. Bassani, "Artificial intelligence and machine learning in spine research," *JOR Spine*, vol. 2, no. 1, Art. no. e1044, Mar. 2019.
- [18] N. Ficarra et al., "Retropubic, laparoscopic, and robot-assisted radical prostatectomy: A systematic review," *European Urology*, vol. 55, no. 5, pp. 1037–1063, May 2009.
- [19] J. Mehrholz et al., "Electromechanical and robot-assisted arm training for improving activities of daily living after stroke," *Cochrane Database of Systematic Reviews*, vol. 2018, no. 9, Art. no. CD006876, Sep. 2018.
- [20] A. M. Okamura, "Haptic feedback in robot-assisted minimally invasive surgery," *Current Opinion in Urology*, vol. 19, no. 1, pp. 102–107, Jan. 2009.
- [21] C. Childers and J. D. Maggard-Gibbons, "Understanding costs of care in the operating room," *JAMA Surgery*, vol. 153, no. 4, Art. no. e176233, Apr. 2018.
- [22] M. U. Scherer, "Regulating artificial intelligence systems: Risks, challenges, competencies, and strategies," *Harvard Journal of Law and Technology*, vol. 29, no. 2, pp. 353–400, Spring 2016.
- [23] B. Sharkey and A. Sharkey, "Granny and the robots: Ethical issues in robot care for the elderly," *Ethics and Information Technology*, vol. 14, no. 1, pp. 27–40, Mar. 2012.
- [24] S. Fusco et al., "An integrated microrobotic platform for on-demand, targeted therapeutic interventions," *Advanced Materials*, vol. 26, no. 6, pp. 952–957, Feb. 2014.
- [25] P. Liang et al., "The first clinical study of using 5G networks to operate a robotic system in telesurgery," *Surgical Endoscopy*, vol. 35, pp. 5344–5350, Sep. 2021.
- [26] A. Donati et al., "Long-term training with a brain-machine interface-based gait protocol induces partial neurological recovery in paraplegic patients," *Scientific Reports*, vol. 6, Art. no. 30383, Aug. 2016.